

EX-1

```
mysql> desc student;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	YES		NULL	
Name	varchar(15)	YES		NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
MobileNo	bigint	YES		NULL	
City	varchar(15)	YES		NULL	

6 rows in set (0.02 sec)

```
mysql> desc faculty;
```

Field	Type	Null	Key	Default	Extra
FacNo	varchar(4)	YES		NULL	
FacName	varchar(15)	YES		NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
DOJ	date	YES		NULL	
MobileNo	bigint	YES		NULL	

6 rows in set (0.00 sec)

```
mysql> desc department;
```

Field	Type	Null	Key	Default	Extra
DeptNo	varchar(4)	YES		NULL	
DeptName	varchar(15)	YES		NULL	
DeptHead	varchar(4)	YES		NULL	

3 rows in set (0.00 sec)

```
mysql> desc course;
```

Field	Type	Null	Key	Default	Extra
CourseNo	varchar(3)	YES		NULL	
CourseDesc	varchar(14)	YES		NULL	
CourseType	char(1)	YES		NULL	
SemNo	char(1)	YES		NULL	
HallNo	varchar(4)	YES		NULL	
FacNo	varchar(4)	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
mysql> desc faculty;
```

Field	Type	Null	Key	Default	Extra
FacNo	varchar(4)	YES		NULL	
FacName	varchar(15)	YES		NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
DOJ	date	YES		NULL	
MobileNo	bigint	YES		NULL	
Dept	varchar(10)	YES		NULL	

```
7 rows in set (0.00 sec)
```

EX-2

```
mysql> desc student;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
Name	varchar(15)	YES		NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
MobileNo	bigint	YES		NULL	
City	varchar(15)	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
mysql> desc faculty;
```

Field	Type	Null	Key	Default	Extra
FacNo	varchar(4)	NO	PRI	NULL	
FacName	varchar(15)	YES		NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
DOJ	date	YES		NULL	
MobileNo	bigint	YES		NULL	
Dept	varchar(10)	YES		NULL	

```
7 rows in set (0.00 sec)
```

```
mysql> desc department;
```

Field	Type	Null	Key	Default	Extra
DeptNo	varchar(4)	NO	PRI	NULL	
DeptName	varchar(15)	YES		NULL	
DeptHead	varchar(4)	YES		NULL	

```
3 rows in set (0.01 sec)
```

```
mysql> desc course;
```

Field	Type	Null	Key	Default	Extra
CourseNo	varchar(3)	NO	PRI	NULL	
CourseDesc	varchar(14)	YES		NULL	
CourseType	char(1)	YES		NULL	
SemNo	char(1)	YES		NULL	
HallNo	varchar(4)	YES		NULL	
FacNo	varchar(4)	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
mysql> desc student;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
Name	varchar(15)	YES	UNI	NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
MobileNo	bigint	YES		NULL	
City	varchar(15)	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
mysql> ALTER TABLE student1 modify address int not null;  
Query OK, 0 rows affected (0.07 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc student1;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
S1Name	varchar(15)	YES		NULL	
Age	char(2)	YES	UNI	NULL	
DOB	int	YES		NULL	
address	int	NO		NULL	

```
5 rows in set (0.00 sec)
```

EX-3

```
mysql> select * from student;
```

RegNo	Name	Gender	DOB	MobileNo	City
101	Arun	M	2004-05-10	9876543210	Chennai
102	Bala	M	2004-08-15	9876543211	Madurai
103	Divya	F	2005-01-20	9876543212	Salem
104	Kavin	M	2004-11-12	9876543213	Coimbatore
105	Nisha	F	2005-03-18	9876543214	Trichy

5 rows in set (0.00 sec)

```
mysql> select * from faculty;
```

FacNo	FacName	Gender	DOB	DOJ	MobileNo	Dept
F001	Ramesh	M	1980-04-15	2010-06-20	9876500001	CSE
F002	Priya	F	1985-08-10	2012-07-15	9876500002	IT
F003	Suresh	M	1978-12-25	2008-01-10	9876500003	ECE
F004	Anitha	F	1983-03-18	2015-09-01	9876500004	EEE
F005	Vijay	M	1982-11-05	2011-05-12	9876500005	MECH

5 rows in set (0.00 sec)

```
mysql> select * from department;
```

DeptNo	DeptName	DeptHead
D001	CSE	F001
D002	IT	F002
D003	ECE	F003
D004	EEE	F004
D005	MECH	F005

5 rows in set (0.00 sec)

```
mysql> select * from course;
```

CourseNo	CourseDesc	CourseType	SemNo	HallNo	FacNo
C01	DBMS	T	3	H101	F001
C02	JAVA	T	4	H102	F002
C03	OS	T	5	H103	F003
C04	CN	T	6	H104	F004
C05	AI	T	6	H105	F005

```
5 rows in set (0.00 sec)
```

EX-4

```
mysql> select * from student;
```

RegNo	Name	Gender	DOB	MobileNo	City
101	Arun Kumar	M	2004-05-10	9876543210	Chennai
102	Bala	M	2004-08-15	9876543211	Madurai
103	Divya	F	2005-01-20	9876543212	Salem
104	Kavin	M	2004-11-12	9876543213	Coimbatore
105	Nisha	F	2005-03-18	9876543214	Trichy

```
5 rows in set (0.00 sec)
```

```
mysql> select * from faculty;
```

FacNo	FacName	Gender	DOB	DOJ	MobileNo	Dept	Resigned
F001	Ramesh	M	1980-04-15	2010-06-20	9876500001	CSE	N
F003	Suresh	M	1978-12-25	2008-01-10	9876500003	ECE	N
F004	Anitha	F	1983-03-18	2015-09-01	9876500004	EEE	N

```
3 rows in set (0.00 sec)
```

```
mysql> select * from faculty;
```

FacNo	FacName	Gender	DOB	DOJ	MobileNo	Dept	Resigned	Age
F001	Ramesh	M	1980-04-15	2010-06-20	9876500001	CSE	N	59
F003	Suresh	M	1978-12-25	2008-01-10	9876500003	ECE	N	NULL
F004	Anitha	F	1983-03-18	2015-09-01	9876500004	EEE	N	NULL

```
3 rows in set (0.00 sec)
```

```
mysql> select * from faculty;
```

FacNo	FacName	Gender	DOB	DOJ	MobileNo	Dept	Resigned	Age
F001	Ramesh	M	1980-04-15	2010-06-20	9876500001	CSE	N	59
F004	Anitha	F	1983-03-18	2015-09-01	9876500004	EEE	N	45

```
2 rows in set (0.00 sec)
```

EX-5

```
mysql> SELECT RegNo, Name, DOB
-> FROM STUDENT;
```

RegNo	Name	DOB
101	Arun Kumar	2004-05-10
102	Bala	2004-08-15
103	Divya	2005-01-20
104	Kavin	2004-11-12
105	Nisha	2005-03-18

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM STUDENT
-> WHERE Gender='F';
```

RegNo	Name	Gender	DOB	MobileNo	City
103	Divya	F	2005-01-20	9876543212	Salem
105	Nisha	F	2005-03-18	9876543214	Trichy

```
2 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM STUDENT
-> WHERE CourseNo='C01';
```

RegNo	Name	Gender	DOB	MobileNo	City	CourseNo
101	Arun Kumar	M	2004-05-10	9876543210	Chennai	C01
102	Bala	M	2004-08-15	9876543211	Madurai	C01
105	Nisha	F	2005-03-18	9876543214	Trichy	C01

3 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM FACULTY
-> WHERE DOJ < '2014-11-01';
```

FacNo	FacName	Gender	DOB	DOJ	MobileNo	Dept	Resigned	Age
F001	Ramesh	M	1980-04-15	2010-06-20	9876500001	CSE	N	59

1 row in set (0.00 sec)

```
mysql> SELECT *
-> FROM COURSE
-> WHERE HallNo <> 'H101';
```

CourseNo	CourseDesc	CourseType	SemNo	HallNo	FacNo
C02	JAVA	T	4	H102	F002
C03	OS	T	5	H103	F003
C04	CN	T	6	H104	F004
C05	AI	T	6	H105	F005

4 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM STUDENT
-> WHERE Name LIKE '%sh';
```

RegNo	Name	Gender	DOB	MobileNo	City	CourseNo
102	Ramesh	M	2004-08-15	9876543211	Madurai	C01

1 row in set (0.00 sec)

```
mysql> SELECT *
-> FROM STUDENT
-> WHERE Name LIKE '%sh%';
```

RegNo	Name	Gender	DOB	MobileNo	City	CourseNo
102	Ramesh	M	2004-08-15	9876543211	Madurai	C01
105	Nisha	F	2005-03-18	9876543214	Trichy	C01

2 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM STUDENT
-> WHERE City LIKE '%Sal%';
```

RegNo	Name	Gender	DOB	MobileNo	City	CourseNo
103	Divya	F	2005-01-20	9876543212	Salem	C02

1 row in set (0.00 sec)

```
mysql> SELECT *
-> FROM STUDENT
-> WHERE Name NOT LIKE '_____';
```

RegNo	Name	Gender	DOB	MobileNo	City	CourseNo
101	Arun Kumar	M	2004-05-10	9876543210	Chennai	C01
103	Divya	F	2005-01-20	9876543212	Salem	C02
104	Kavin	M	2004-11-12	9876543213	Coimbatore	C03
105	Nisha	F	2005-03-18	9876543214	Trichy	C01

4 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM STUDENT
-> WHERE Name LIKE '%am%';
```

RegNo	Name	Gender	DOB	MobileNo	City	CourseNo
102	Ramesh	M	2004-08-15	9876543211	Madurai	C01

1 row in set (0.00 sec)

EX-6

```
mysql> SELECT CourseType
      -> FROM COURSE
      -> WHERE CourseDesc IN ('DBMS','JAVA');
+-----+
| CourseType |
+-----+
| T          |
| T          |
+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT CourseNo
      -> FROM StudentScores
      -> WHERE Score BETWEEN 50 AND 80;
+-----+
| CourseNo |
+-----+
| C01      |
| C01      |
| C02      |
| C02      |
+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT AVG(Score)
      -> FROM StudentScores
      -> WHERE CourseNo='C02';
+-----+
| AVG(Score) |
+-----+
| 75.0000    |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT MAX(Score), MIN(Score)
-> FROM StudentScores
-> WHERE CourseNo='C02';
```

MAX(Score)	MIN(Score)
85	65

```
1 row in set (0.00 sec)
```

```
mysql> SELECT CourseNo,
-> MAX(Score),
-> MIN(Score),
-> AVG(Score)
-> FROM StudentScores
-> GROUP BY CourseNo;
```

CourseNo	MAX(Score)	MIN(Score)	AVG(Score)
C01	90	70	80.0000
C02	85	65	75.0000

```
2 rows in set (0.00 sec)
```

```
mysql> SELECT CourseName,
-> AVG(Score)
-> FROM StudentScores
-> GROUP BY CourseName;
```

CourseName	AVG(Score)
DBMS	80.0000
JAVA	75.0000

```
2 rows in set (0.00 sec)
```

```
mysql> SELECT SUM(Score)
-> FROM StudentScores;
+-----+
| SUM(Score) |
+-----+
|          465 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT CourseDesc,
-> COUNT(StudentNo)
-> FROM StudentScores
-> GROUP BY CourseDesc;
+-----+-----+
| CourseDesc | COUNT(StudentNo) |
+-----+-----+
| CSE        | 3                |
| Programming | 3                |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT StudentNo,
-> COUNT(CourseNo)
-> FROM StudentScores
-> GROUP BY StudentNo;
+-----+-----+
| StudentNo | COUNT(CourseNo) |
+-----+-----+
| 101       | 2                |
| 102       | 1                |
| 105       | 1                |
| 103       | 1                |
| 104       | 1                |
+-----+-----+
5 rows in set (0.00 sec)
```

EX-7

```
mysql> SELECT CourseDesc,  
->          COUNT(StudentNo) AS NoOfStudents  
-> FROM StudentScores  
-> GROUP BY CourseDesc;
```

CourseDesc	NoOfStudents
CSE	3
Programming	3

2 rows in set (0.03 sec)

```
mysql> SELECT StudentNo,  
->          COUNT(CourseNo) AS NoOfCourses  
-> FROM StudentScores  
-> GROUP BY StudentNo;
```

StudentNo	NoOfCourses
101	2
102	1
105	1
103	1
104	1

5 rows in set (0.00 sec)

```
mysql> SELECT Name, Gender, MobileNo  
-> FROM STUDENT  
-> ORDER BY RegNo ASC;
```

Name	Gender	MobileNo
Arun Kumar	M	9876543210
Ramesh	M	9876543211
Divya	F	9876543212
Kavin	M	9876543213
Nisha	F	9876543214

5 rows in set (0.01 sec)

```
mysql> SELECT *  
-> FROM FACULTY  
-> ORDER BY DOB ASC;
```

FacNo	FacName	Gender	DOB	DOJ	MobileNo	Dept	Resigned	Age
F001	Ramesh	M	1980-04-15	2010-06-20	9876500001	CSE	N	59
F004	Anitha	F	1983-03-18	2015-09-01	9876500004	EEE	N	45

2 rows in set (0.02 sec)

EX-8

```
mysql> SELECT *
-> FROM StudentScores
-> WHERE Score >
-> (
->     SELECT AVG(Score)
->     FROM StudentScores
-> );
```

StudentNo	CourseNo	CourseName	CourseDesc	Score
102	C01	DBMS	CSE	80
105	C01	DBMS	CSE	90
104	C02	JAVA	Programming	85

3 rows in set (0.01 sec)

```
mysql> SELECT StudentNo,
->     COUNT(*) AS NoOfTests
-> FROM StudentScores
-> GROUP BY StudentNo
-> HAVING COUNT(*) > 1;
```

StudentNo	NoOfTests
101	2

1 row in set (0.00 sec)

```
mysql> SELECT FacName, DOJ
-> FROM FACULTY
-> WHERE DOJ =
-> (
->     SELECT MAX(DOJ)
->     FROM FACULTY
-> );
```

```
+-----+-----+
| FacName | DOJ          |
+-----+-----+
| Anitha  | 2015-09-01  |
+-----+-----+
1 row in set (0.01 sec)
```

```
mysql> SELECT CourseNo,
->         StudentNo,
->         Score
-> FROM StudentScores s1
-> WHERE Score >
-> (
->     SELECT AVG(Score)
->     FROM StudentScores s2
->     WHERE s1.CourseNo = s2.CourseNo
-> );
```

```
+-----+-----+-----+
| CourseNo | StudentNo | Score |
+-----+-----+-----+
| C01      | 105       | 90    |
| C02      | 104       | 85    |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

EX-9

```
mysql> SELECT F.FacName,  
->          D.DeptName  
-> FROM FACULTY F  
-> INNER JOIN DEPARTMENT D  
-> ON F.Dept = D.DeptName;
```

FacName	DeptName
Ramesh	CSE
Anitha	EEE

2 rows in set (0.03 sec)

```
mysql> SELECT S.Name,  
->          S.CourseNo  
-> FROM STUDENT S  
-> LEFT JOIN StudentScores SS  
-> ON S.RegNo = SS.StudentNo  
-> WHERE SS.StudentNo IS NULL;
```

Name	CourseNo
Karthik	C02

1 row in set (0.00 sec)

```
mysql> SELECT C.CourseDesc  
-> FROM COURSE C  
-> INNER JOIN FACULTY F  
-> ON C.FacNo = F.FacNo  
-> WHERE F.FacName='Kamal'  
-> AND F.Dept='ECE';
```

```
+-----+  
| CourseDesc |  
+-----+  
| DBMS       |  
+-----+
```

```
1 row in set (0.00 sec)
```

EX-10

```
mysql> SELECT * FROM Student_View;
```

RegNo	Name	City
101	Arun Kumar	Chennai
102	Ramesh	Madurai
103	Divya	Salem
104	Kavin	Coimbatore
105	Nisha	Trichy
106	Karthik	Chennai

```
6 rows in set (0.06 sec)
```

```
mysql> SELECT * FROM Faculty_View;
+-----+-----+-----+
| FacNo | FacName | Dept |
+-----+-----+-----+
| F001  | Ratnam  | Sales |
| F004  | Anitha  | EEE   |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SHOW INDEX FROM STUDENT;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
student	0	PRIMARY	1	RegNo	A	5				BTREE			YES	NULL
student	0	uk_name	1	Name	A	5				BTREE			YES	NULL
student	1	idx_student_name	1	Name	A	6				BTREE			YES	NULL

3 rows in set (0.05 sec)

```
mysql> SHOW INDEX FROM FACULTY;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
faculty	0	PRIMARY	1	FacNo	A	3	NULL	NULL	YES	BTREE			YES	NULL
faculty	1	idx_faculty_name	1	FacName	A	2	NULL	NULL	YES	BTREE			YES	NULL

2 rows in set (0.01 sec)